

Seat No.: _____

Paper Code: OS-601-SET4

MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION**B.TECH. DEGREE SEMESTER EXAMINATION 2026****Subject: Operating Systems & System Architecture****Day & Date: Sep 23, 2026****Duration: 3 Hours****Max. Marks: 70****Instructions to Candidates:**

1. All questions are compulsory subject to internal choices.
2. Figures to the right indicate full marks for each question.
3. Draw neat, labeled diagrams wherever necessary.
4. Assume suitable data if required and state the assumptions clearly.
5. Use of non-programmable scientific calculator is permitted.

SECTION – I (OBJECTIVE & MULTIPLE CHOICE QUESTIONS)**[14 Marks]***Q.1 Select and write the most appropriate alternative from the options: (1 Mark each)*

1. Which of the following CPU scheduling algorithms is strictly non-preemptive? **[1]**
 (A) Round Robin (B) FCFS
 (C) SRTF (D) Preemptive Priority
2. A thread within a process shares which of the following resources with other threads? **[1]**
 (A) Stack and Registers (B) Code, Data and Open Files
 (C) Program Counter (D) CPU Registers
3. The Banker's algorithm is used in operating systems for which primary purpose? **[1]**
 (A) Deadlock Recovery (B) Deadlock Avoidance
 (C) Deadlock Prevention (D) Mutual Exclusion
4. Which page replacement algorithm suffers from Belady's anomaly? **[1]**
 (A) Optimal (OPT) (B) FIFO
 (C) LRU (D) LFU

SECTION – II (SHORT ANSWER & ANALYTICAL QUESTIONS)**[28 Marks]***Q.2 Answer any FOUR of the following questions: (7 Marks each)*

- (a) Explain the four necessary conditions for Deadlock occurrence and how prevention eliminates them. **[7]**
- (b) Describe the working of Round Robin CPU scheduling with an example and draw a Gantt Chart. **[7]**
- (c) Compare Monolithic Kernel and Microkernel architecture with neat block diagrams. **[7]**
- (d) Explain the concept of Virtual Memory and how Page Fault Handling operates in demand paging. **[7]**

SECTION – III (COMPREHENSIVE & DESIGN PROBLEMS)**[28 Marks]***Q.3 Answer the following design and analytical questions: (7 Marks each)*

- (a) Explain the Critical Section Problem and illustrate how Peterson's Algorithm achieves Mutual Exclusion. **[7]**

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 Explain the Producer-Consumer problem and solve it using Counting & Binary Semaphores. **[7]**
- (b) Consider 5 processes and 3 resource types. Explain the Safety Algorithm in Banker's Algorithm with data structures. **[7]**

***** END OF QUESTION PAPER *****

